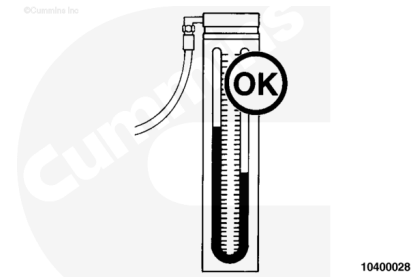


Trainings ▾

General Information

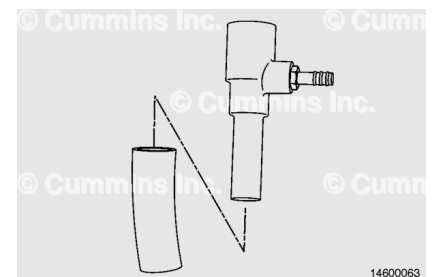
Excessive blowby indicates an air compressor, turbocharger, or engine malfunction that allows combustion gasses or air to enter the crankcase and build a pressure higher than normal.

This procedure describes how to measure crankcase pressure in single and dual bank crankcase breather equipped models.



Note : This procedure requires the use of two Blowby Tools, Part Number 3822571.

It is essential that two 10.3 mm [0.406 in] Orifice Blowby Tools are used. One barbed connection must be connected to a manometer and one barbed connection plugged or capped off.



Failure to do so will cause the blowby readings to be inaccurate.

Measure

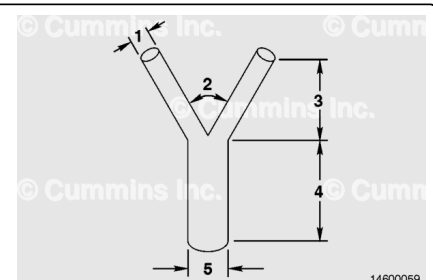
The following is for a single bank crankcase breather blowby tool installation.

Fabricate a “Y” piece to the following specifications:

1. 25.4 mm [1 in] (internal diameter)
2. 60 degrees
3. 101.6 mm [4 in]
4. 101.6 mm [4 in]
5. 50.8 mm [2 in] (internal diameter).

Pipe thickness - 1.5 mm [0.059 in].

Note : The “Y” piece, must have a 60 degree angle at the vee (2) and be symmetrical to make sure blowby



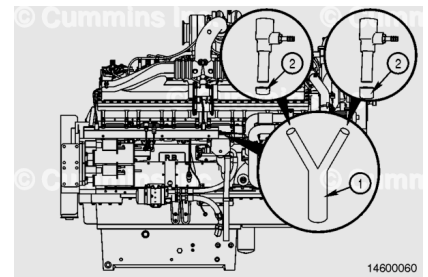
gases are split equally.

If applicable, remove the 50.4 mm [2 in] flexible crankcase breather pipe.

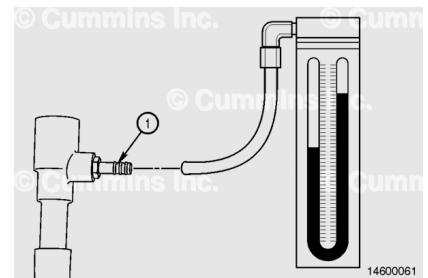
Use a short length of 50.4 mm [2 in] internal diameter hose to connect directly from the crankcase breather body to the 50.4 mm [2 in] "Y" pipe (1).

Use a suitable pipe (2) and attach two 10.3 mm [0.406 in] orifice blowby tools, Part Number 3822571, to each arm of the "Y" pipe.

Note : It is essential that two 10.3 mm [0.406 in] Orifice Blowby Tools are used. One barbed connection must be connected to a manometer and one barbed connection plugged or capped off. Failure to do so will cause the blowby reading to be inaccurate.



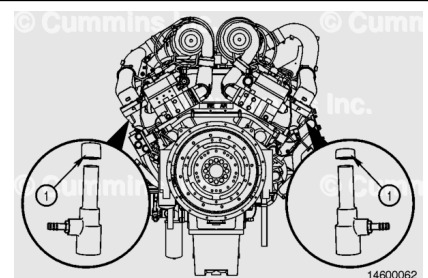
Attach a water manometer, Part Number ST-1111-3 or equivalent, to the unplugged barbed connection of the first blowby measurement tool (1). Make sure the barbed connection on the second blowby measurement tool is plugged or capped off.



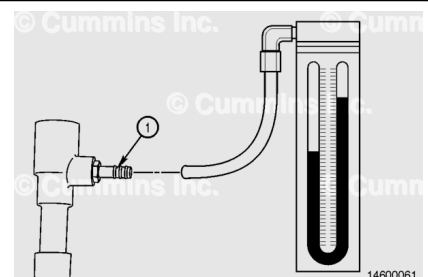
The following information is for a dual bank crankcase breather blowby tool installation.

Use a short length of suitable hose (1) to attach each 10.3 mm [0.406 in] orifice blowby tool, Part Number 3822571, to each of the crankcase breathers.

Note : It is essential that two 10.3 mm [0.406 in] Orifice Blowby Tools are used. One barbed connection must be open and one barbed connection plugged or capped off. Failure to do so will cause the blowby reading to be inaccurate.



Attach a water manometer, Part Number ST-1111-3 or equivalent, to the unplugged barbed connection of the first blowby measurement tool (1). Make sure the barbed connection on the second blowby measurement tool is plugged or capped off.

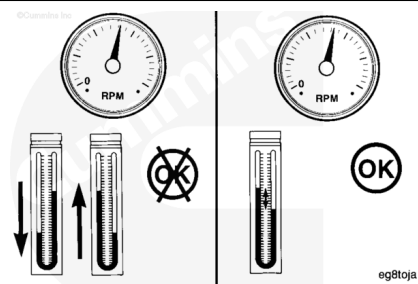


Test

Run the engine until it reaches normal operating temperatures and pressures.

Operate the engine at rated RPM and full load (wide-open throttle) until a steady reading is obtained on the manometer.

A sudden increase in the blowby readings indicates a problem. A gradual increase over time is normal (due to wear on the internal engine components).



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Compare the blowby readings to previous readings on the engine.

If the previous readings for the engine are **not** available, use the information in the table below.

QSK45 and QSK60 Blowby Limits

Use two 10.3 mm [0.406 in] diameter orifice Blowby Measurement Tools at Rated rpm and load.

Engine Family	New Engine Limits	Rebuilt Engine Limits	Used Engine Limits
QSK45 and QSK60	2.49 kPa [10 in H ₂ O]	2.49 kPa [10 in H ₂ O]	5.47 kPa [22 in H ₂ O]

Last Modified: 20-Jul-2023
